

Project Istanbul

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1. Project Overview and Scope

Istanbul is the most populous city in Europe. Consequently, in the event of a disaster such as an earthquake, both the scale of potential damage and the demand for assistance would be substantial.

This project focuses on Istanbul, a city widely expected to experience a major earthquake and considerable damage. The objective is not only to analyze buildings and residential areas in terms of structural risk and provide district-level assessments, but also to examine the perceptions of residents living in Istanbul and to compare these findings with the analytical results.

To achieve this, four different datasets were utilized, and a variety of graphs were generated and interpreted in detail to derive meaningful insights.

2. Data

2.1 Data Source

[Deprem Senaryosu Analiz Sonuçları](#)

[İstanbul Çevresinde Gerçekleşen Depremler 2019-2020](#)

[2019-2024 Yılları Belediye Nüfusları](#)

[2017 Mahalle Bazlı Bina Sayıları](#)

[VDYM İlçe Bazında Hanelerdeki Kentsel Dönüşüm Fikri Bireysel Öngörülleri](#) is a dataset about the opinions of households about the idea of urban transformation on a district basis.

All this data taken from [İstanbul Büyükşehir Belediyesi Açık Veri Portalı](#)

District boundary geometries used for mapping were obtained from [GADM version 4.1](#).

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Data quality checks after preprocessing

dataset	rows	columns	missing_values	duplicate_rows
Earthquake scenario	959	15	0	0
Building stock	959	9	0	0
Observed earthquakes	349	9	0	0
District population	39	8	0	0
Urban transformation	39	5	0	0

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2.2 General Information About Data

[Deprem Senaryosu Analiz Sonuçları](#) is a dataset that contains the results of an earthquake scenario analysis. Its columns mainly include information on the number of buildings by damage level, the number of people by vital status, and pipe damage.

[İstanbul Çevresinde Gerçekleşen Depremler 2019-2020](#) is a dataset on earthquakes that occurred around Istanbul. Each row represents a single earthquake, and the columns provide various details about these events.

[2019-2024 Yılları Belediye Nüfusları](#) is a dataset that provides population information for each municipality.

[2017 Mahalle Bazlı Bina Sayıları](#) is a dataset about buildings at the neighbourhood level. Each row represents a neighbourhood, and the columns mainly include information about the construction year of buildings and the number of floors.

2.3 Reason of Choice

Istanbul is widely regarded as a city facing a very high earthquake risk, with the potential for severe loss of life and property. Experts frequently warn in the media that a major earthquake, estimated to range between magnitudes 7.2 and 7.6, could strike the city. I chose to focus on this highly relevant issue in order to inform the public through a comprehensive analysis, raise awareness, and encourage precautionary measures that could help minimize the impact of such a disaster.

2.4 Preprocessing

Five datasets were downloaded from the sources listed in Section 2.1 and imported from their original Excel files using `readxl`. All character columns were converted to UTF-8, repeated whitespace was removed, and empty strings were represented as missing values. District names were standardized in Turkish uppercase and the word “Belediyesi” was removed from the population dataset to create a common `ilce_adi` key. The historical district name “Eyüp” was also standardized as “Eyüpsultan” to match the more recent datasets.

The earthquake timestamps were converted to UTC date-time values, while magnitude, depth, latitude, and longitude were explicitly checked as numeric variables. Fully duplicated rows were removed, and the district lists were validated across the earthquake scenario, building stock, population, and urban transformation datasets. The cleaned datasets were then stored in `istanbul_deprem_verileri.RData`.

2.5 District-Level Data Integration

The earthquake scenario and building stock datasets originally contain one row per neighbourhood. Their count variables were summed to produce one row for each district. The population and urban transformation datasets already contain one row per district, so only their variable names and district keys were standardized.

The observed earthquake dataset cannot be assigned directly to Istanbul districts because its coordinates cover a much wider region, including Western Türkiye, the Aegean Sea, the Black Sea, Bulgaria, and surrounding areas. Assigning these events to Istanbul districts would therefore be misleading. Instead, this fifth dataset contributes regional context variables, including the total number of observed earthquakes, the number recorded within 100 km of central Istanbul, mean and maximum magnitude, median depth, and the observation period. These contextual values are identical for all district rows and must not be interpreted as district-specific earthquake counts.

The resulting `ilce_birlesik` table contains exactly one row for each of Istanbul’s 39 districts. It combines projected earthquake consequences, building characteristics, annual population, urban transformation opinions, and regional observed-earthquake context. It also includes derived variables for the proportion of pre-1980 buildings, the proportion reporting a need for urban transformation, and population change between 2019 and 2024.

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Validation of the district-level integrated dataset

item	value
Districts	39
Variables	42
Missing values	0
Duplicate district rows	0

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Selected variables for the ten districts with the highest projected loss of life

ilce_adi	can_kaybi_sayisi	1980_onesi	nufus_2024	kentsel_donusum_evet_orani	istanbula_100km_deprem
BAHÇELİEVLER	1633	3892	560086	29.2%	
KÜÇÜKÇEKMECE	1515	7535	789033	22.4%	
FATİH	1484	31899	354472	31.4%	

ilce_adi	can_kaybi_sayisi	1980_onesi	nufus_2024	kentsel_donusum_evet_orani	istanbula_100km_deprem
BAĞCILAR	1179	6072	713594	29.8%	
BAKIRKÖY	1046	4869	219893	33.3%	
ESENYURT	1003	499	988369	8.0%	
GÜNGÖREN	754	2513	264831	44.7%	
ZEYTİNBURNU	668	2594	278344	35.1%	
ESENLER	638	6554	423625	43.4%	
BEYLİKDÜZÜ	527	41	415290	8.5%	

3. Analysis

3.1 Exploratory Data Analysis

The exploratory analysis first examines the structure and quality of the integrated district-level dataset. Raw counts are considered together with population- and building-adjusted rates because districts differ considerably in population and building stock. The observed-earthquake variables are excluded from district comparisons because they describe the same regional context for every district.

3.1.1 Variable Dictionary

The following table documents the principal variables used in the district-level analysis. Count variables from the scenario and building datasets were aggregated by summing neighbourhood values within each district.

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Dictionary of the principal district-level variables

variable	source	meaning	unit
ilce_adi	All district sources	Standardized district name	Category
can_kaybi_sayisi	Earthquake scenario	Projected loss of life	People
cok_agir_hasarli_bina_sayisi	Earthquake scenario	Projected number of severely damaged buildings	Buildings
agir_hasarli_bina_sayisi	Earthquake scenario	Projected number of heavily damaged buildings	Buildings
gecici_barinma	Earthquake scenario	Projected temporary shelter requirement	People
1980_onesi	Building stock	Buildings constructed before 1980	Buildings

variable	source	meaning	unit
toplam_bina	Building stock	Total buildings across the three construction periods	Buildings
eski_bina_orani	Derived	Share of buildings constructed before 1980	Proportion
nufus_2024	Population	District population in 2024	People
nufus_degisim_orani_2019_2024	Derived	Population change between 2019 and 2024	Proportion
kentsel_donusum_evet_orani	Urban transformation	Share reporting a need for urban transformation	Proportion
can_kaybi_100bin	Derived	Projected loss of life per 100,000 residents	People per 100,000
gecici_barinma_100bin	Derived	Temporary shelter requirement per 100,000 residents	People per 100,000
agir_ve_cok_agir_hasar_orani	Derived	Heavily and severely damaged buildings relative to total buildings	Proportion

3.1.2 Data Quality Profile

The integrated table was checked for missing values, duplicated district records, impossible negative values, and variation across districts. Variables with only one unique value are regional context variables copied from the observed-earthquake dataset; they are retained for documentation but are not used in correlations or district rankings.

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Quality profile of variables used in exploratory analysis

variable	missing	unique	negative
can_kaybi_sayisi	0	38	0
cok_agir_hasarli_bina_sayisi	0	38	0
agir_hasarli_bina_sayisi	0	38	0
gecici_barinma	0	39	0
1980_onesi	0	39	0
toplam_bina	0	39	0
eski_bina_orani	0	39	0
nufus_2024	0	39	0
nufus_degisim_orani_2019_2024	0	39	21
kentsel_donusum_evet_orani	0	39	0
can_kaybi_100bin	0	38	0
gecici_barinma_100bin	0	39	0
agir_ve_cok_agir_hasar_orani	0	39	0

All 39 districts are represented once in the integrated table. No missing values, duplicate district rows, or impossible negative counts were found in the selected variables. The negative values appearing in the population-change variable are valid and indicate population decline rather than a data-quality problem.

3.1.3 Descriptive Statistics

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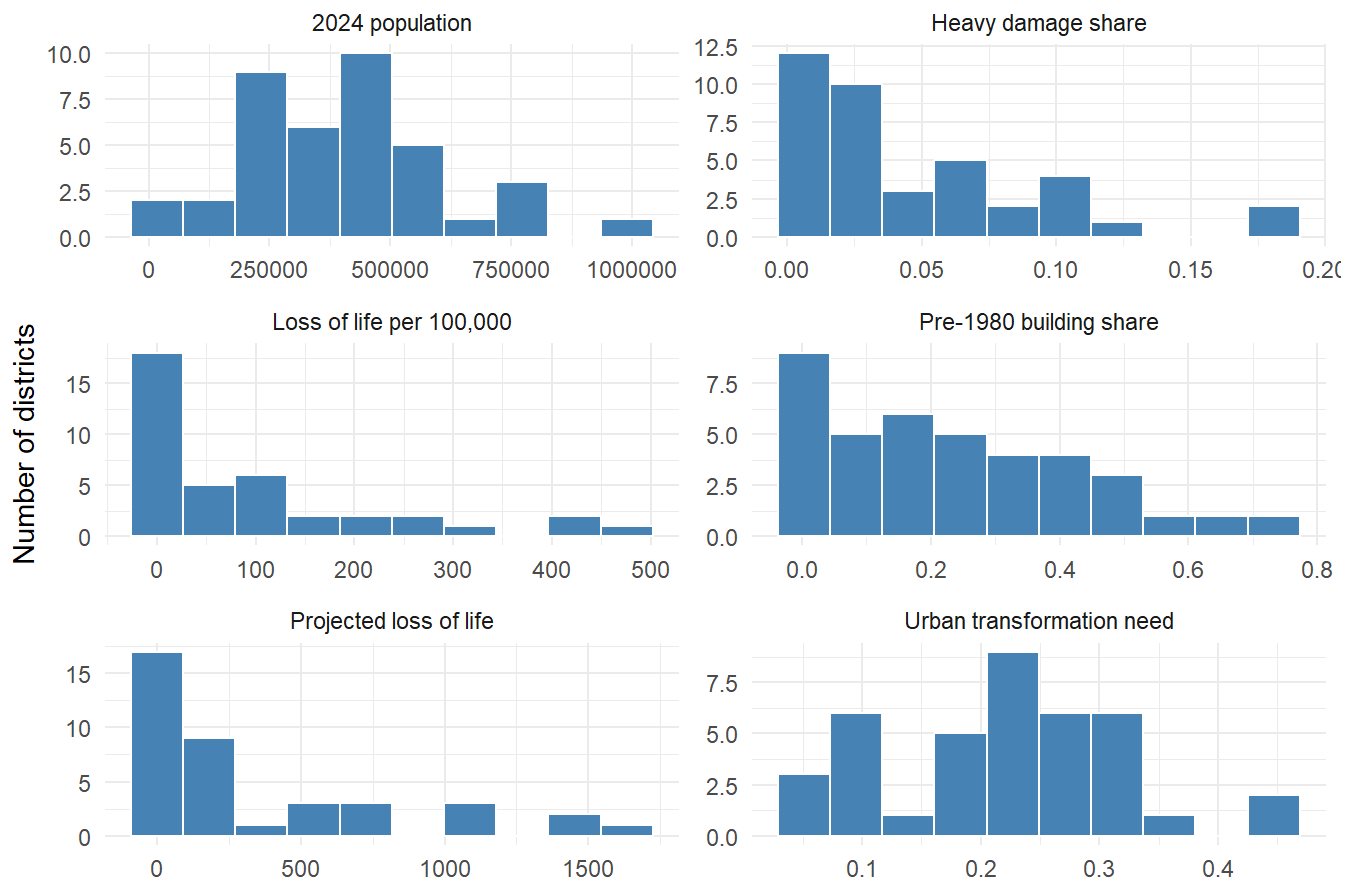
Descriptive statistics across Istanbul's 39 districts							
variable	minimum	q1	median	mean	q3	maximum	sd
can_kaybi_sayisi	0.000	56.000	168.000	362.692	523.500	1633.000	463.851
cok_agir_hasarli_bina_sayisi	16.000	103.500	233.000	346.051	441.500	2083.000	392.188
agir_hasarli_bina_sayisi	106.000	409.500	648.000	880.359	1221.500	3496.000	713.511
gecici_barinma	876.000	10101.500	16067.000	21433.795	27879.500	72774.000	18164.377
1980_onesi	41.000	1640.000	5448.000	6622.282	9906.000	31899.000	6367.936
toplam_bina	6393.000	22114.000	28742.000	29895.795	36524.000	52612.000	11873.513
eski_bina_orani	0.003	0.069	0.188	0.234	0.372	0.732	0.195
nufus_2024	16979.000	266567.000	414866.000	402605.179	507529.000	988369.000	205089.707
nufus_degisim_orani_2019_2024	-0.200	-0.042	-0.005	0.028	0.101	0.298	0.102
kentsel_donusum_evet_orani	0.051	0.150	0.224	0.219	0.289	0.447	0.098
can_kaybi_100bin	0.000	14.593	36.986	100.207	138.752	475.686	129.373
gecici_barinma_100bin	1278.611	2433.872	3822.351	5571.934	8132.106	18346.192	4008.825
agir_ve_cok_agir_hasar_orani	0.006	0.014	0.028	0.048	0.065	0.181	0.045

The substantial difference between the mean and median for projected loss of life indicates a right-skewed distribution: a small number of districts account for particularly high projected consequences. Building age also varies greatly. The median district has approximately 18.8% pre-1980 buildings, whereas the maximum district-level share exceeds 73%.

3.1.4 Distributions

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Distribution of Selected District Indicators



The histograms show that projected losses and damage are concentrated in the upper tail rather than distributed evenly across Istanbul. Population-adjusted loss therefore provides an important complement to raw loss counts, particularly for smaller but structurally vulnerable districts.

3.1.5 Outlier Assessment

Potential outliers were identified using the conventional 1.5 interquartile-range rule. These observations are not removed because they represent real districts and may be substantively important for disaster planning.

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Potential district outliers based on the 1.5 IQR rule

variable	ilce_adi	value
1980_onesi	FATİH	31899.00
agir_hasarli_bina_sayisi	FATİH	3496.00
agir_hasarli_bina_sayisi	KÜÇÜKÇEKMECE	2616.00
agir_ve_cok_agir_hasar_orani	ADALAR	0.18
agir_ve_cok_agir_hasar_orani	BAKIRKÖY	0.17
can_kaybi_100bin	BAKIRKÖY	475.69
can_kaybi_100bin	ADALAR	447.61

variable	ilce_adi	value
can_kaybi_100bin	FATİH	418.65
can_kaybi_sayisi	BAHÇELİEVLER	1633.00
can_kaybi_sayisi	KÜÇÜKÇEKMECE	1515.00
can_kaybi_sayisi	FATİH	1484.00
cok_agir_hasarli_bina_sayisi	FATİH	2083.00
cok_agir_hasarli_bina_sayisi	KÜÇÜKÇEKMECE	1240.00
gecici_barinma	KÜÇÜKÇEKMECE	72774.00
gecici_barinma	ESENYURT	67410.00
gecici_barinma	BAHÇELİEVLER	61199.00
gecici_barinma	BAGCILAR	59513.00
nufus_2024	ESENYURT	988369.00

Fatih repeatedly appears as an outlier in building-age and damage indicators, while Bahçelievler and Küçükçekmece stand out for projected loss of life. Esenyurt is an outlier in population. These observations should be investigated rather than deleted because each reflects a distinct dimension of exposure or vulnerability.

3.1.6 Relationships Between Indicators

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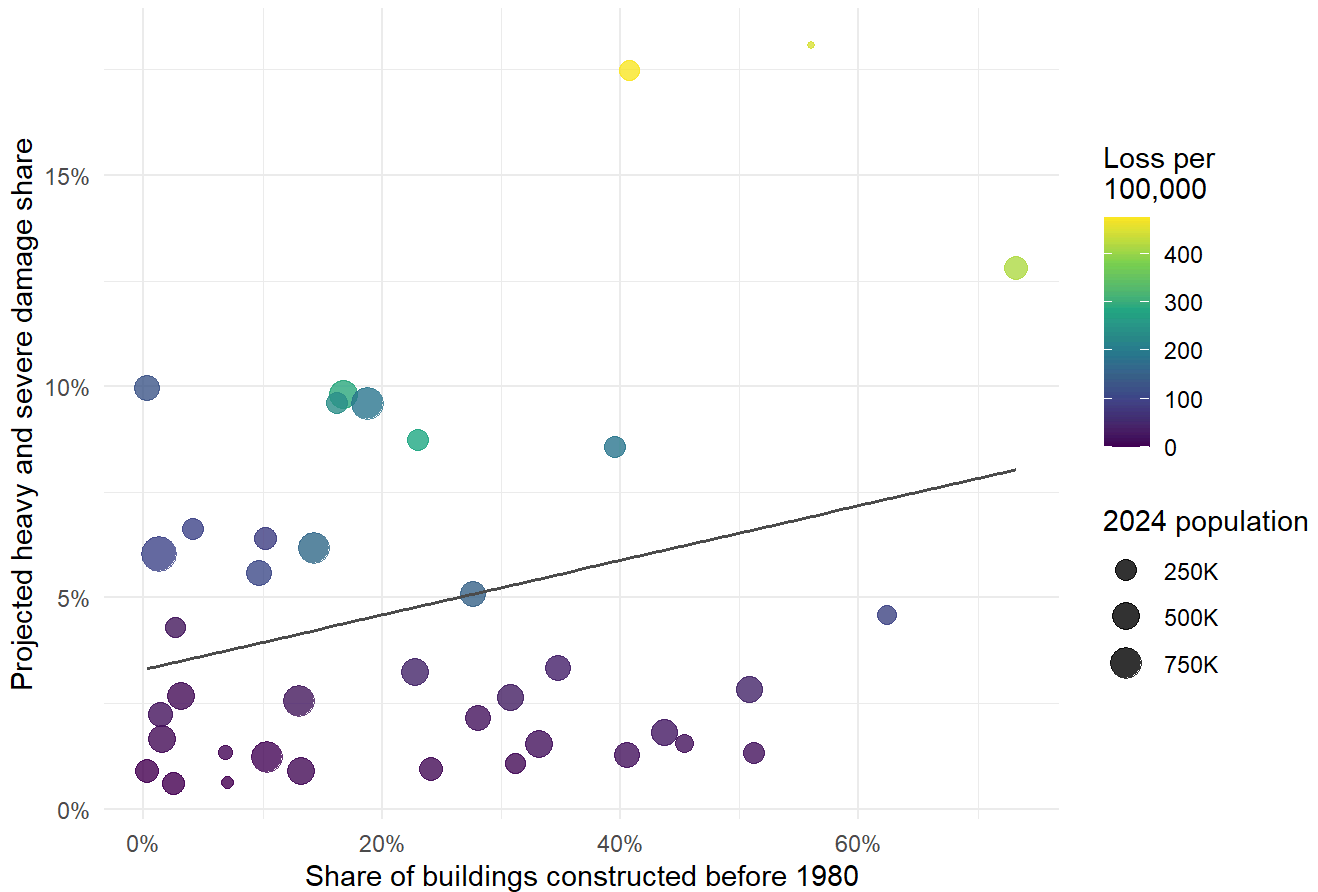
Spearman correlations between selected district indicators

variable	Loss of life	Loss per 100,000	Heavy damage share	Pre-1980 share	Population 2024	Population change	Transformation need
Loss of life	1.00	0.91	0.86	0.18	0.32	-0.47	0.37
Loss per 100,000	0.91	1.00	0.95	0.34	0.02	-0.47	0.36
Heavy damage share	0.86	0.95	1.00	0.16	-0.01	-0.27	0.15
Pre-1980 share	0.18	0.34	0.16	1.00	-0.23	-0.71	0.55
Population 2024	0.32	0.02	-0.01	-0.23	1.00	0.00	0.00
Population change	-0.47	-0.47	-0.27	-0.71	0.00	1.00	-0.76
Transformation need	0.37	0.36	0.15	0.55	0.00	-0.76	1.00

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```
`geom_smooth()` using formula = 'y ~ x'
```


Building Age and Projected Heavy Damage



Spearman correlation is used because several indicators are strongly skewed and the sample consists of only 39 districts. The strongest relationship is between projected heavy-damage share and loss of life per 100,000 residents ($\rho = 0.95$). The pre-1980 building share has a moderate positive relationship with perceived urban transformation need ($\rho = 0.55$), while population change and transformation need have a strong negative relationship ($\rho = -0.76$). These correlations describe association, not causation. The scatter plot adds population and projected loss per 100,000 residents, allowing structural vulnerability, exposure, and consequences to be considered together.

3.1.7 Source-Specific Visualizations

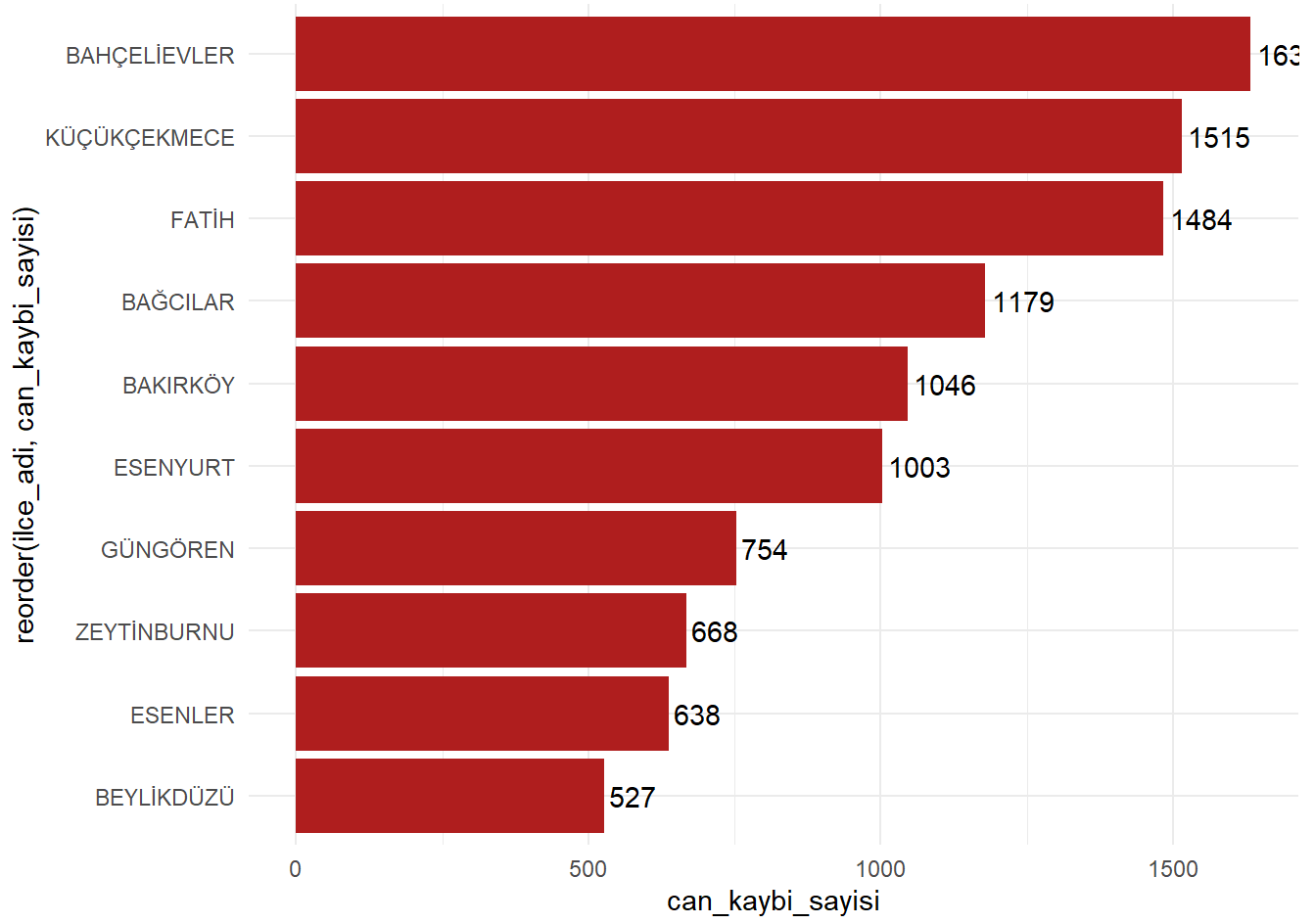
“deprensenaryo” analysis:

The analysis began with the “deprensenaryo” dataset. The top three districts in projected loss of life are Bahçelievler, Küçükçekmece, and Fatih.

Fatih, Küçükçekmece, and Bahçelievler also have the highest projected numbers of severely injured residents.

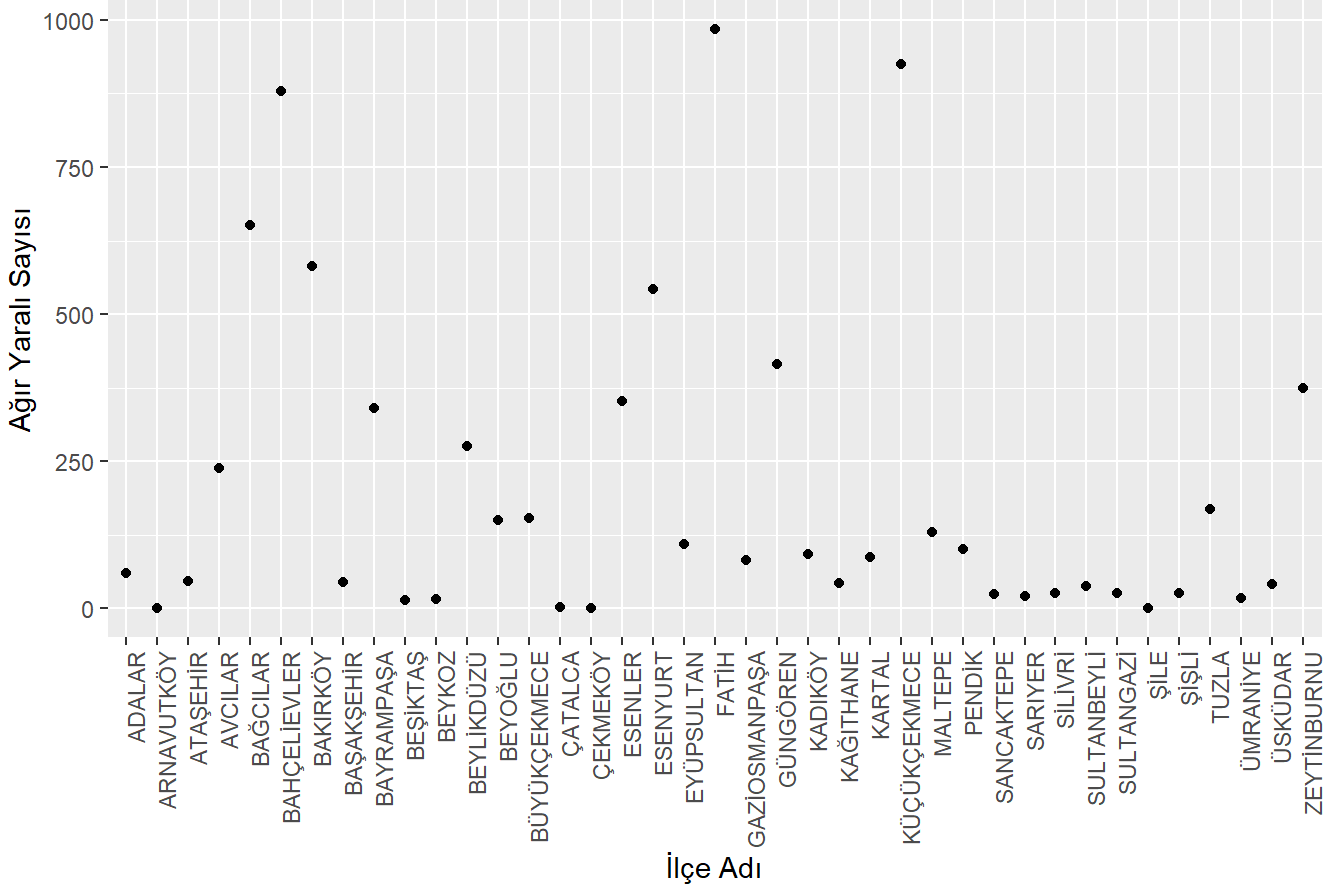
Bahçelievler, Küçükçekmece, and Fatih lead the hospital-treatment and minor-injury projections, while Küçükçekmece, Esenyurt, and Bahçelievler have the highest temporary shelter requirements.

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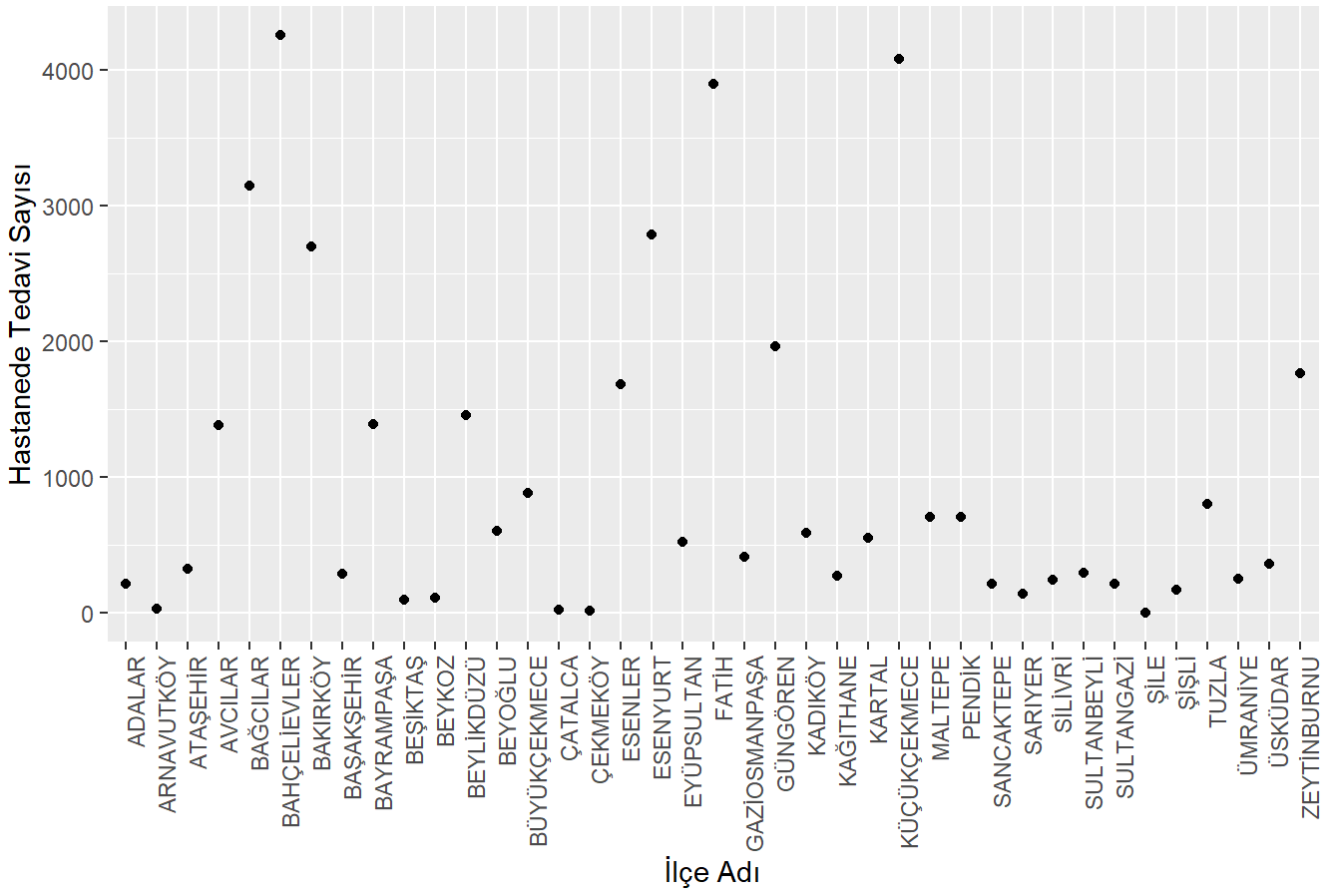
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İlçe Bazında Ağır Yaralı Sayısı Dağılımı



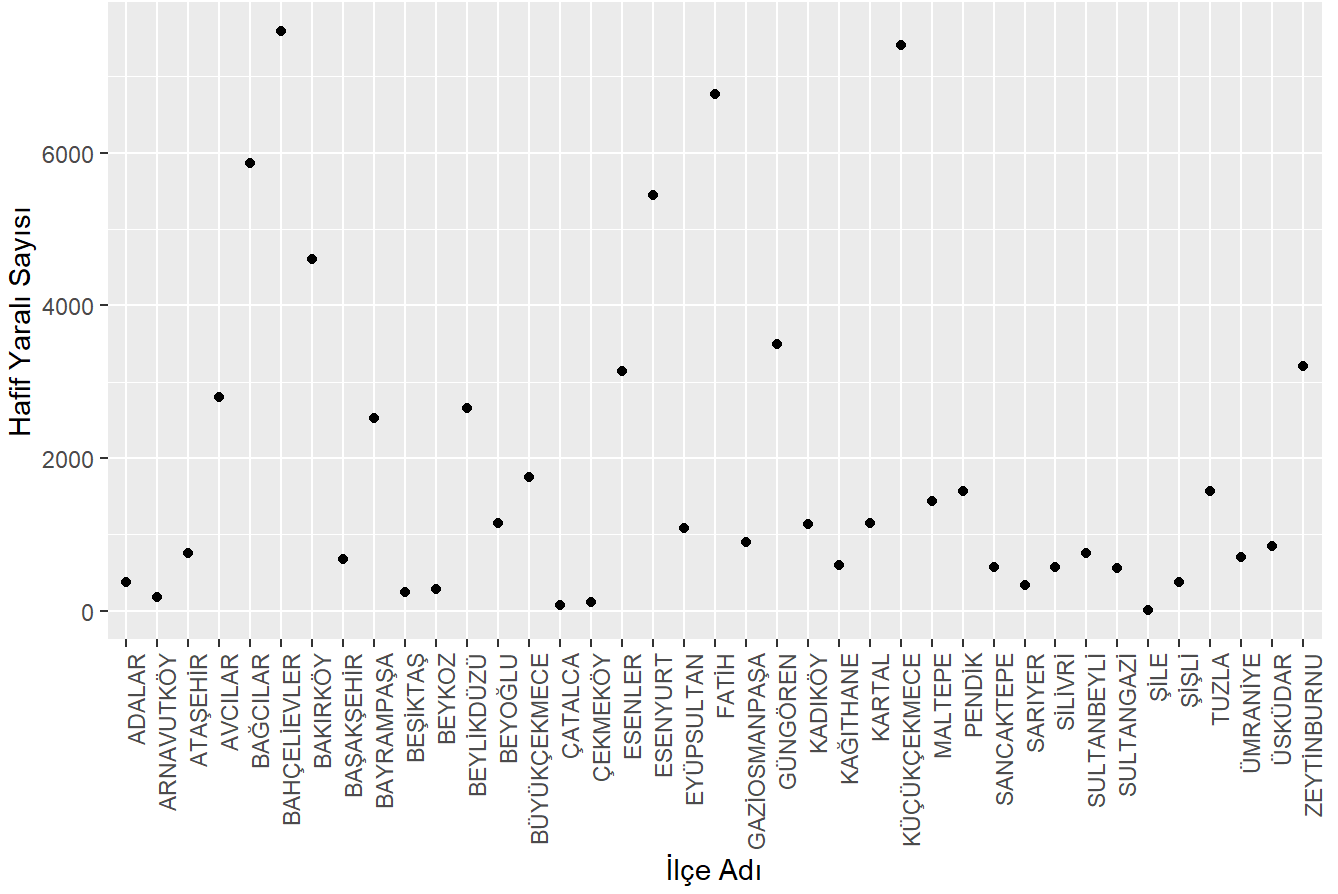
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İlçe Bazında Hastanede Tedavi Sayısı Dağılımı



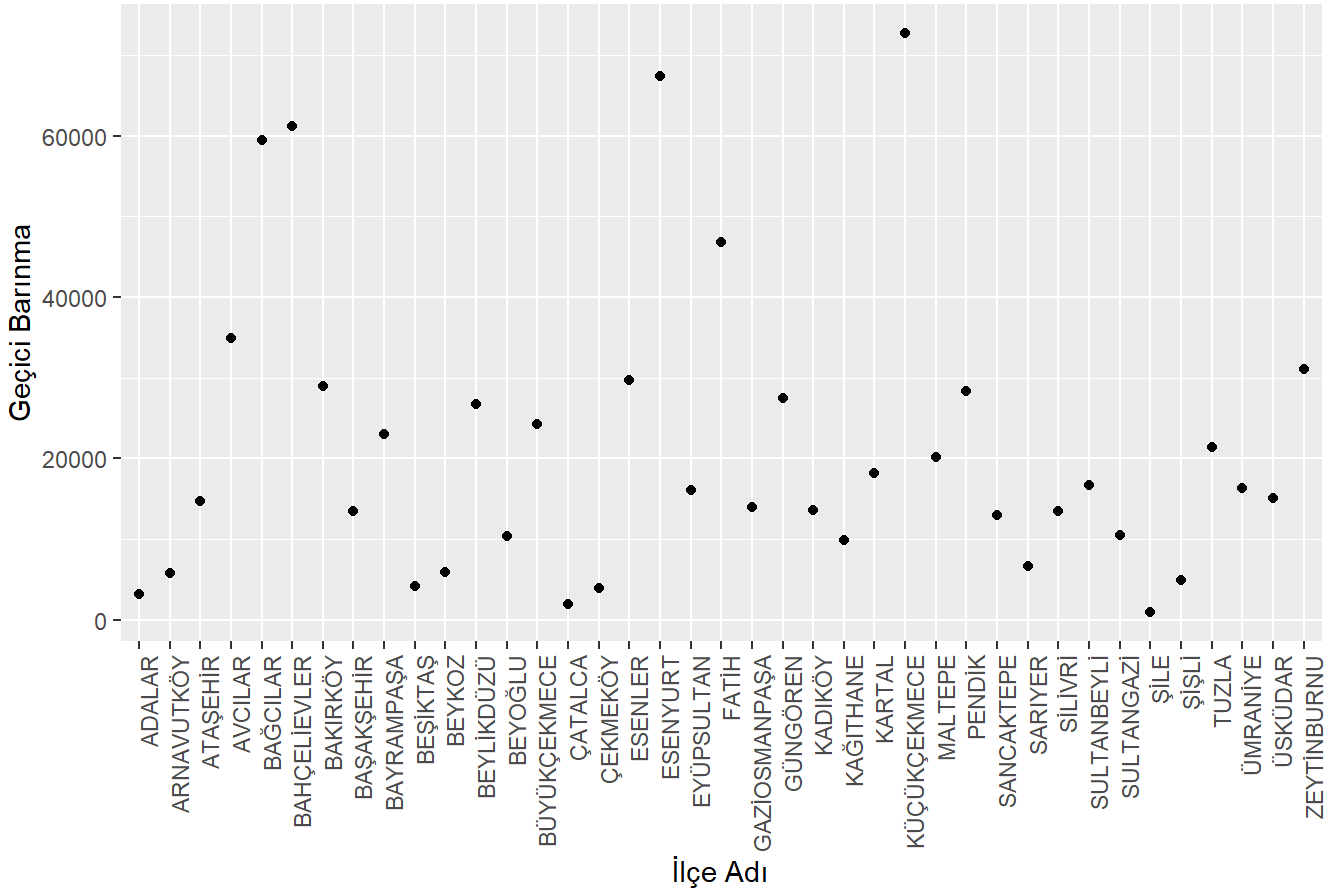
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İlçe Bazında Hafif Yaralı Sayısı Dağılımı



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İlçe Bazında Geçici Barınma Dağılımı

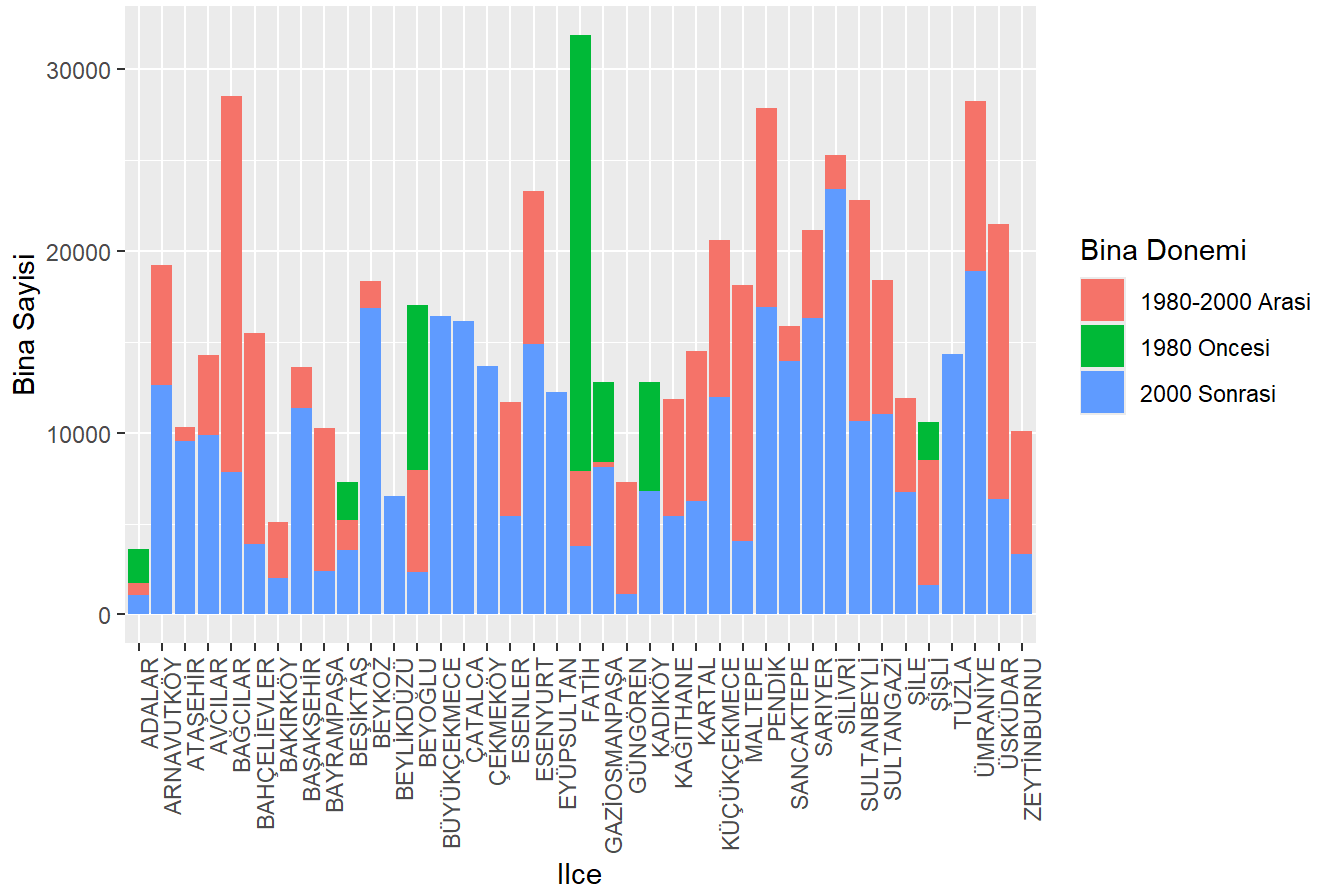


The analysis continued with the “mahbinasayi” dataset, evaluating buildings based on their construction year and number of floors.

Although the data is based on neighborhoods, the analyses have been done based on districts.

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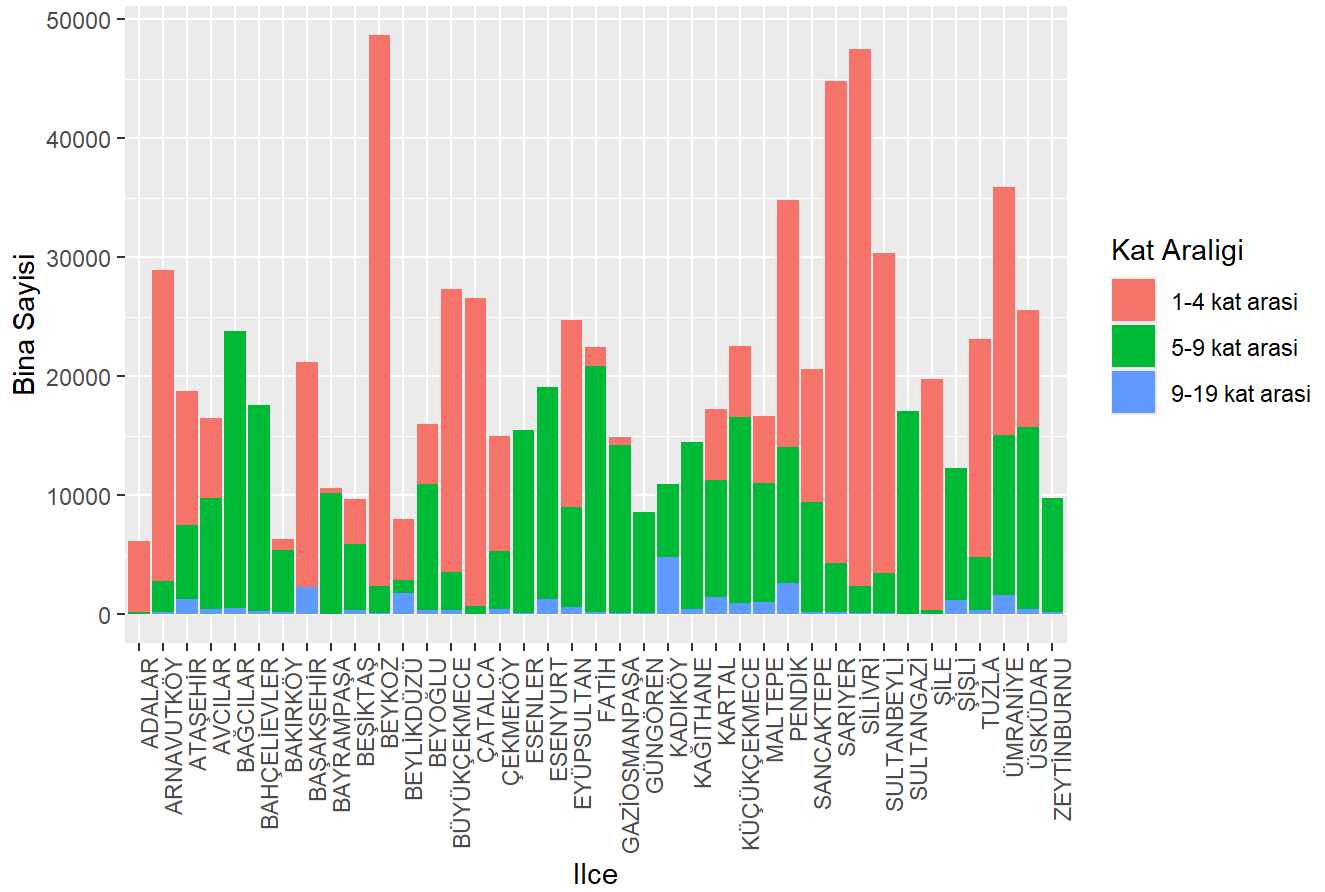
İlcelere Gore Bina Yasi Dagilimi



“ilçe bazında kat sayısına göre bina sayıları”:

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İlcelere Gore Kat Sayisi Dagilimi

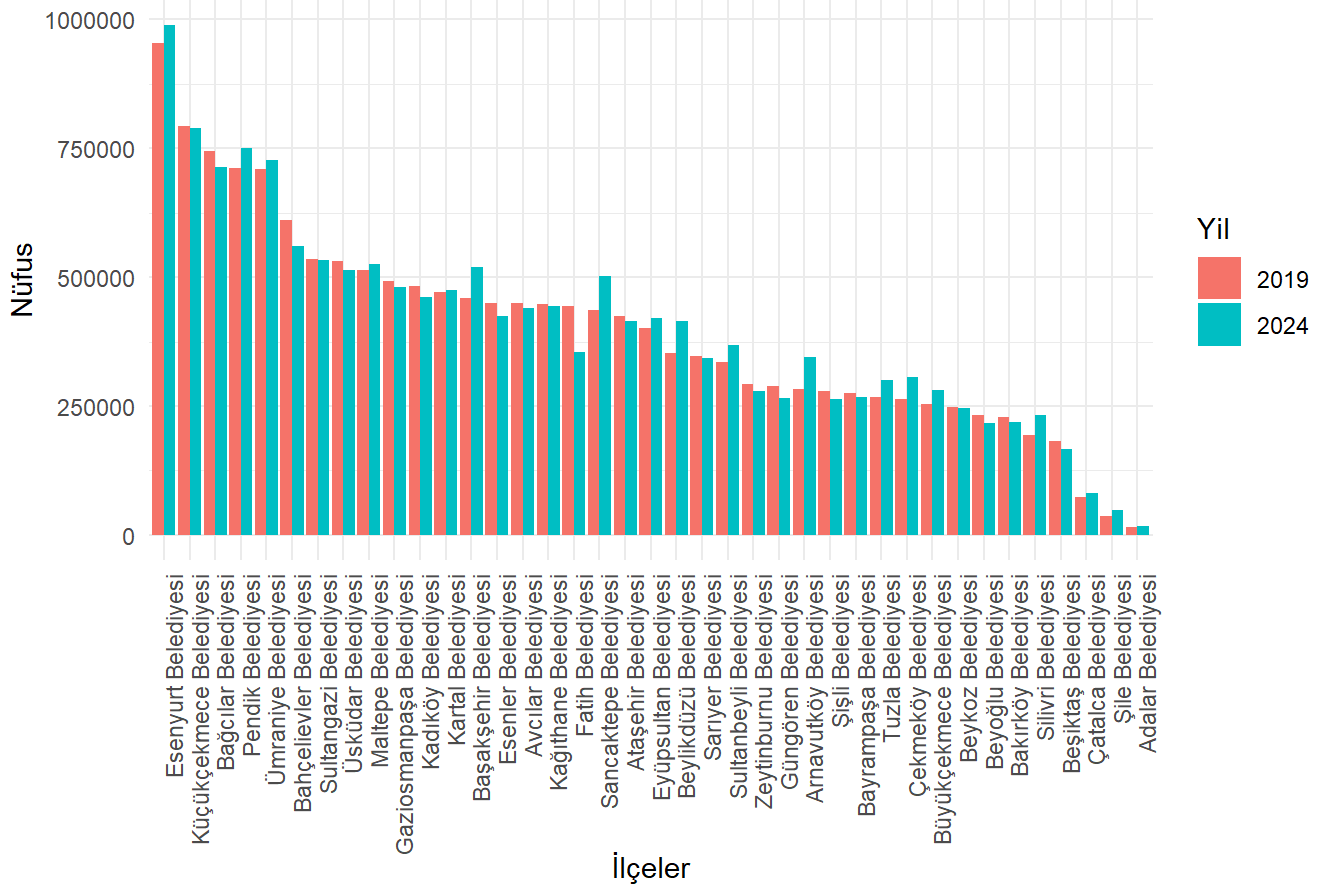


belediyenufus analysis:

In the analysis of the "belediyenufus" dataset, municipalities have been ranked from highest to lowest population and it has been observed that the most crowded municipality is the Esenyurt Municipality.

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2019 ve 2024 İlçe Nüfusları



kentseldonusum analysis:

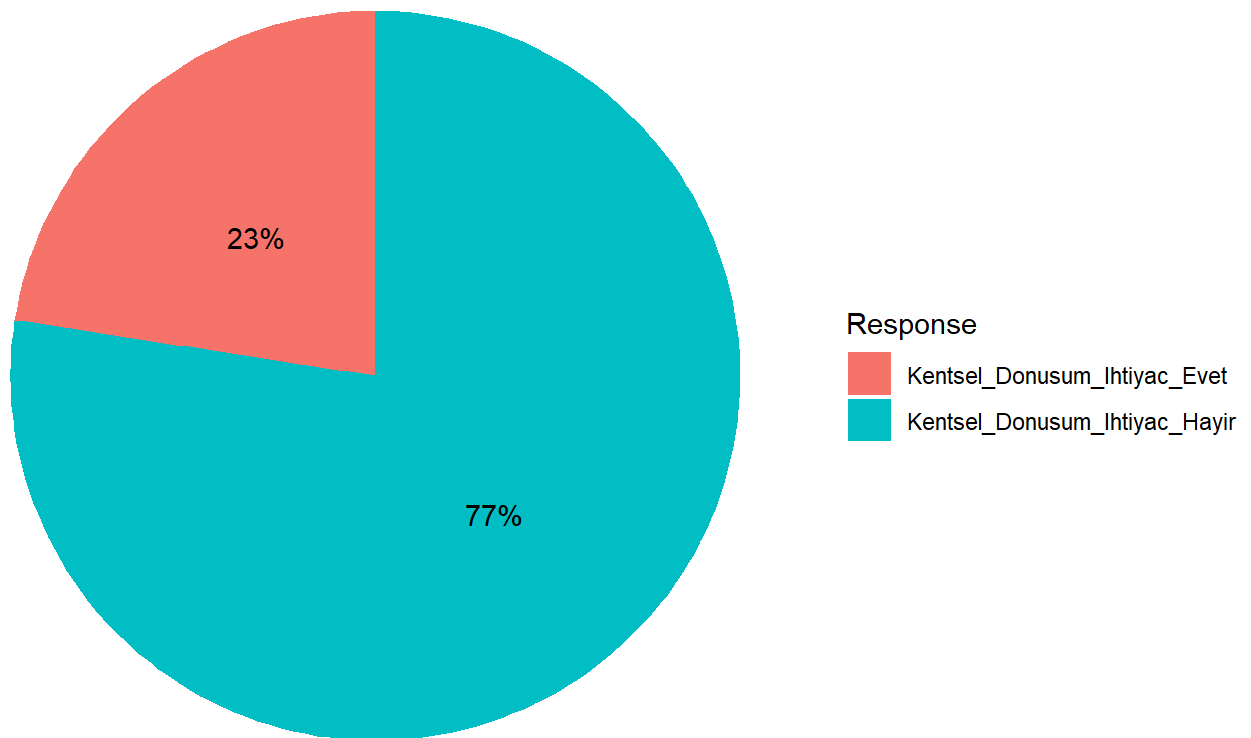
An analysis of individual perceptions regarding the idea of urban transformation of their households have been conducted.

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Kentsel_Donusum_Ihtiyac_Evet	Kentsel_Donusum_Ihtiyac_Hayir
0.2255722	0.7744278

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Need for Urban Transformation

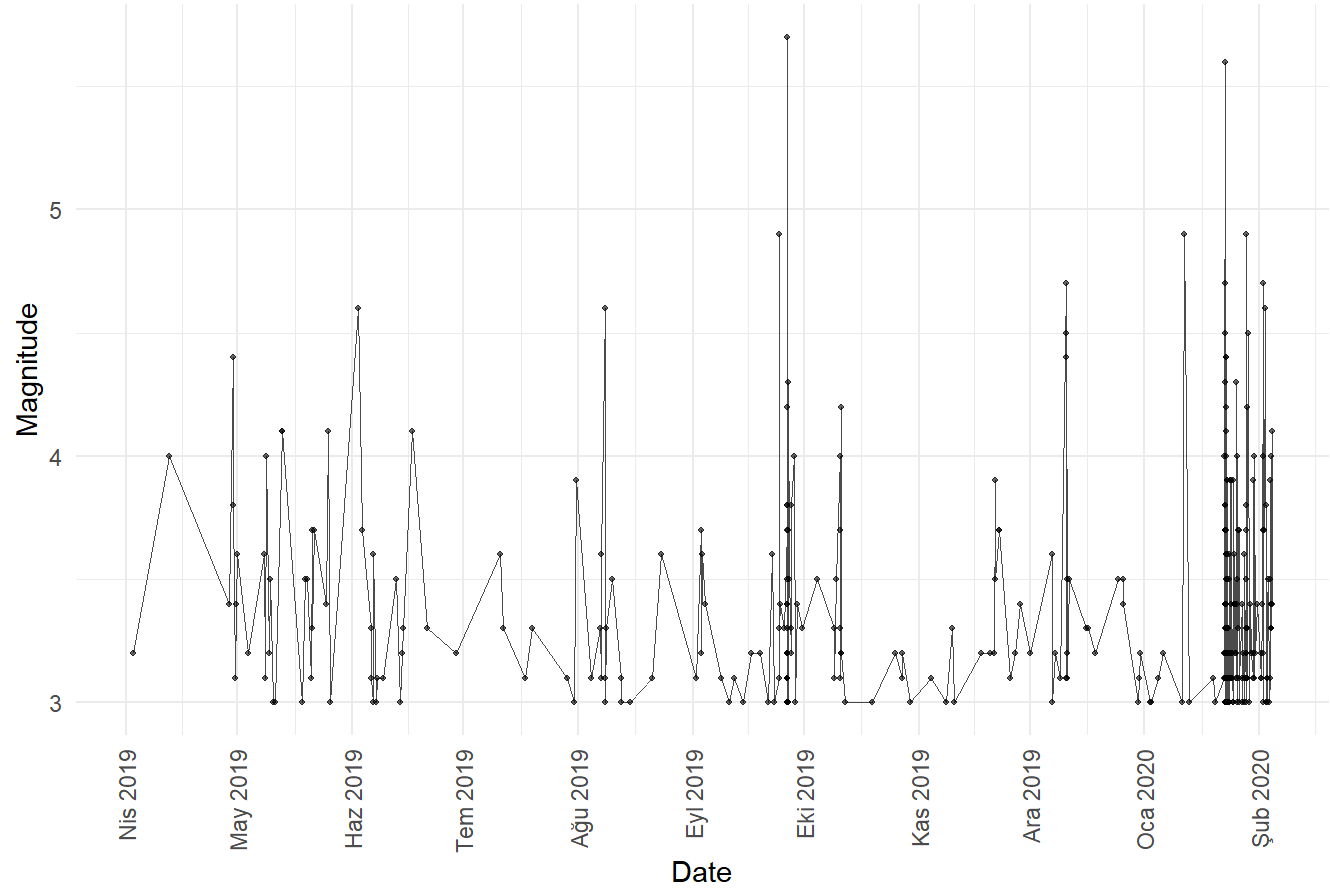


istanbuldepremverisi analysis:

Analyzing this dataset is essential for understanding earthquake activity and assessing potential risks. However, forecasting future earthquake magnitudes remains highly challenging, as earthquakes are inherently random and unpredictable.

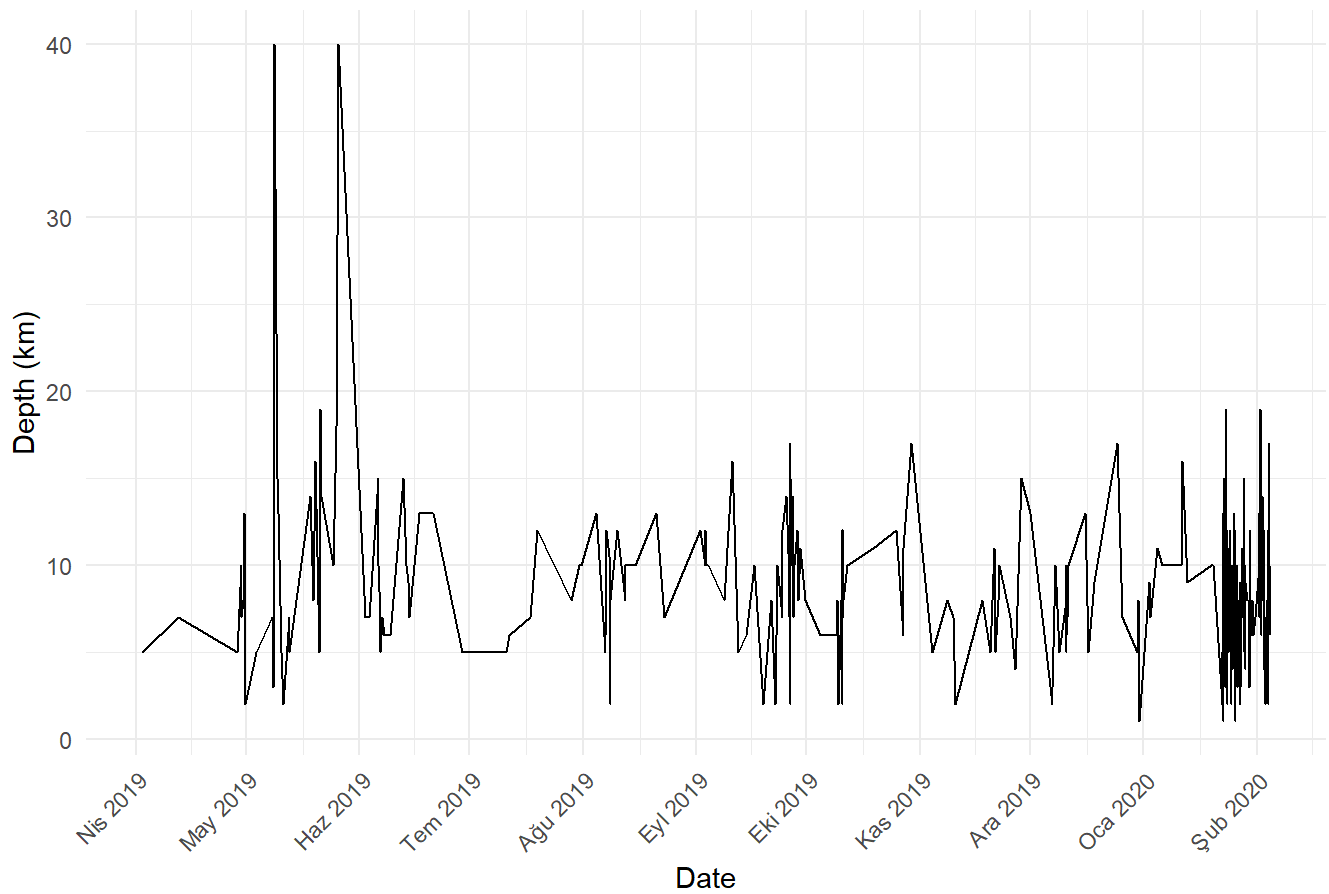
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Magnitudes of Earthquakes Around Istanbul



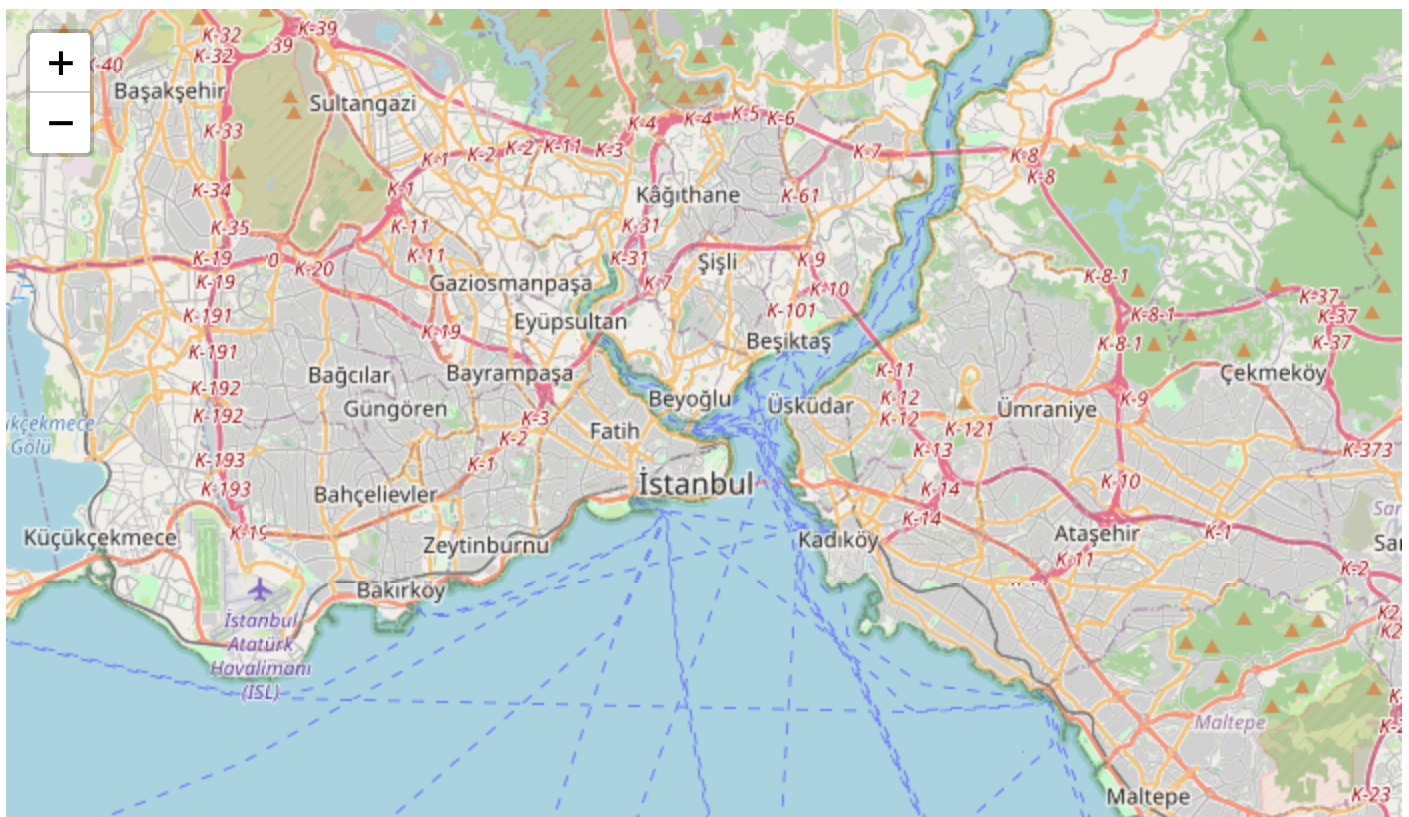
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Depths of Earthquakes Around Istanbul



Display of Earthquakes Occurring Around Istanbul on the Map

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3.2 Normalized District Risk Score

3.2.1 Method

A composite score was developed to compare the relative earthquake risk of Istanbul’s districts. The score does not estimate the probability that an earthquake will occur. Instead, it combines the projected consequences, structural vulnerability, infrastructure and shelter pressure, and the scale of exposed population and buildings under the scenario data.

Each indicator was normalized to the 0–100 range using min–max normalization:

$$z_{ij} = 100 \times \frac{x_{ij} - \min(x_j)}{\max(x_j) - \min(x_j)}$$

where x_{ij} is the value of district i for indicator j . A value of 0 represents the lowest observed value among the 39 districts and 100 represents the highest. Indicators within each dimension were averaged, and the four dimension scores were assigned equal weights. Equal weights were chosen because no expert-derived weighting scheme was available.

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Dimensions and weights of the normalized district risk score

dimension	weight	indicators
Human consequences	25%	Loss of life, severe injuries, and hospital treatment per 100,000 residents
Structural vulnerability	25%	Heavy/severe damage share and pre-1980 building share
Infrastructure and shelter	25%	Temporary shelter and natural gas, drinking-water, and wastewater damage per 100,000 residents
Exposure	25%	2024 population and total building stock

3.2.2 Calculation

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The 15 districts with the highest normalized risk scores

risk_sirasi	ilce_adi	risk_puani	risk_grubu	insani_sonuc_puani	yapisal_kirilganlik_puani	altyapi_barinma_puani
1	ADALAR	71.6	Very high	98.0	88.2	1
2	FATİH	64.2	Very high	83.8	84.9	
3	BAKIRKÖY	56.5	Very high	89.9	76.0	

risk_sirasi	ilce_adi	risk_puani	risk_grubu	insani_sonuc_puani	yapisal_kirilganlik_puani	altyapi_barinma_puani
4	KÜÇÜKÇEKMECE	43.2	Very high	37.8		38.4
5	BAHÇELİEVLER	40.8	Very high	54.8		37.6
6	BAĞCILAR	37.4	Very high	31.6		25.5
7	ESENYURT	33.9	Very high	19.6		16.2
8	BAYRAMPAŞA	33.7	Very high	38.9		49.7
9	GÜNGÖREN	33.2	High	53.8		38.8
10	ZEYTİNBURNU	33.0	High	45.8		36.6
11	BÜYÜKÇEKMECE	30.9	High	20.5		19.8
12	BEYOĞLU	29.3	High	20.8		53.9
13	PENDİK	28.6	High	5.5		14.3
14	SİLİVRİ	28.2	High	5.6		12.2
15	ESENLER	28.0	High	28.6		31.6

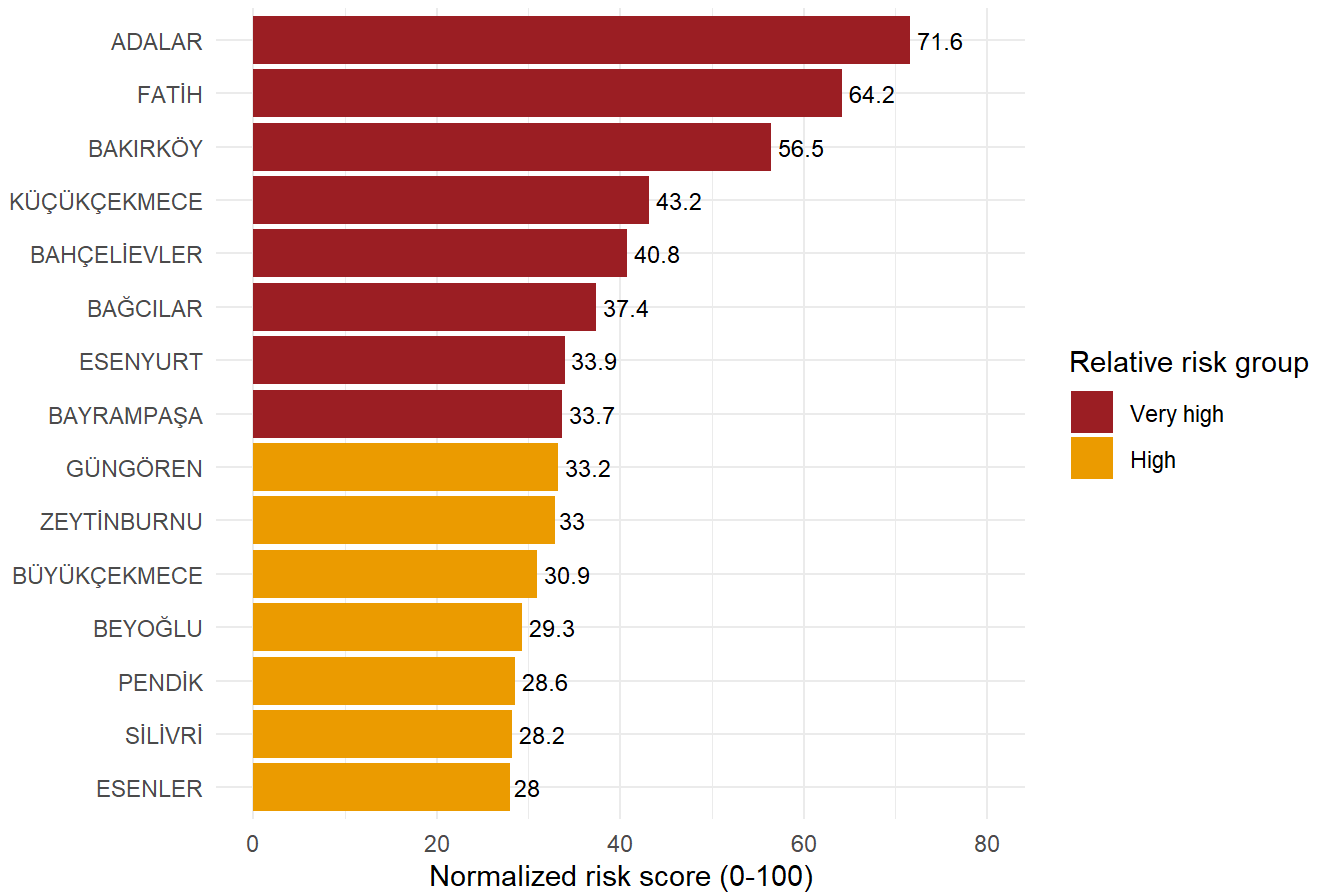
Risk groups are relative quintiles rather than absolute safety thresholds. A district classified as “very low” therefore has lower modeled risk than the other Istanbul districts, but it should not be interpreted as having no earthquake risk.

3.2.3 Ranking and Risk Components

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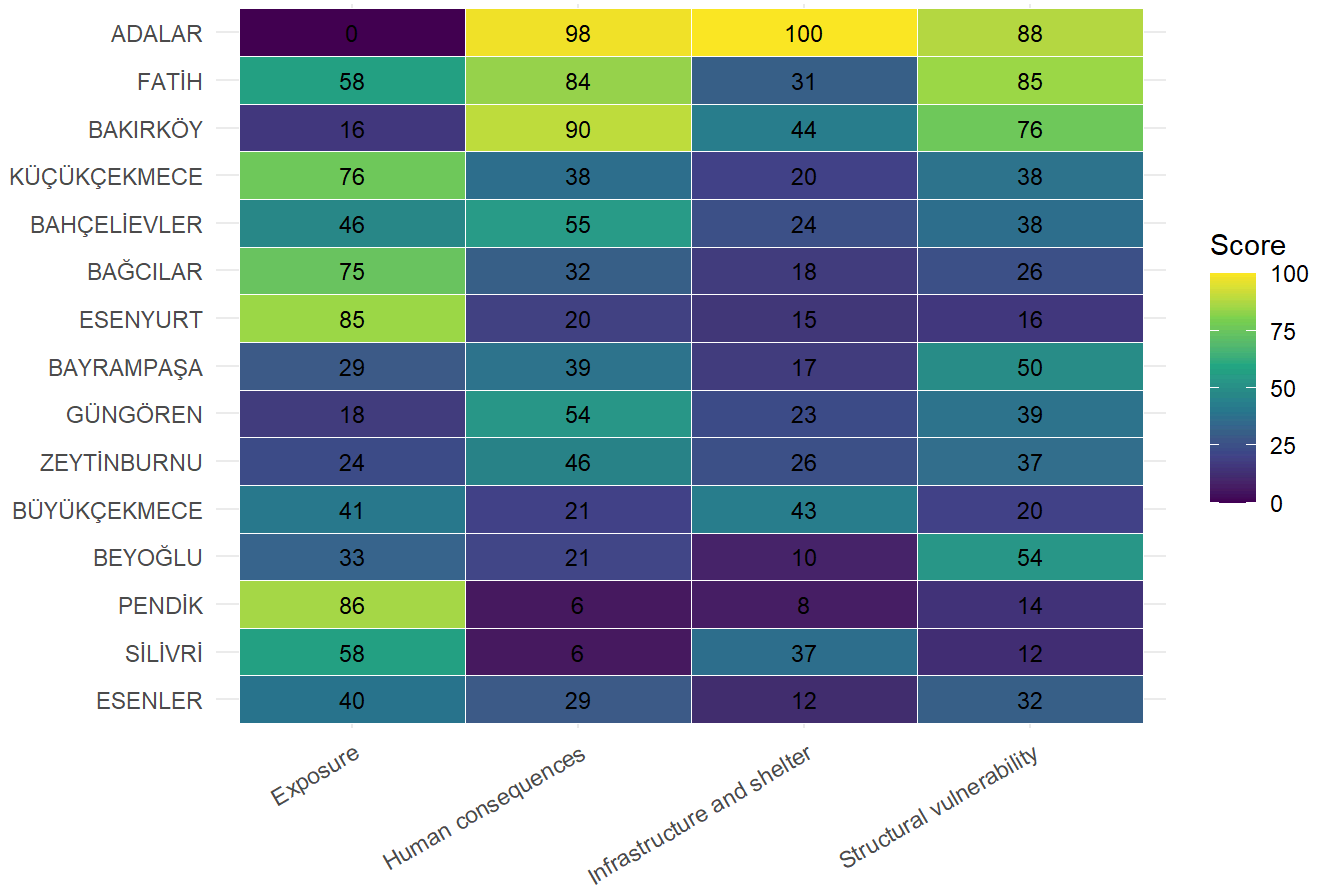


Districts with the Highest Composite Risk Scores



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Risk Components of the 15 Highest-Ranked Districts



3.2.4 Weight Sensitivity

An alternative score was calculated by assigning 35% to human consequences, 30% to structural vulnerability, 20% to infrastructure and shelter, and 15% to exposure. This alternative reflects a stronger priority on preventing loss of life and structural damage.

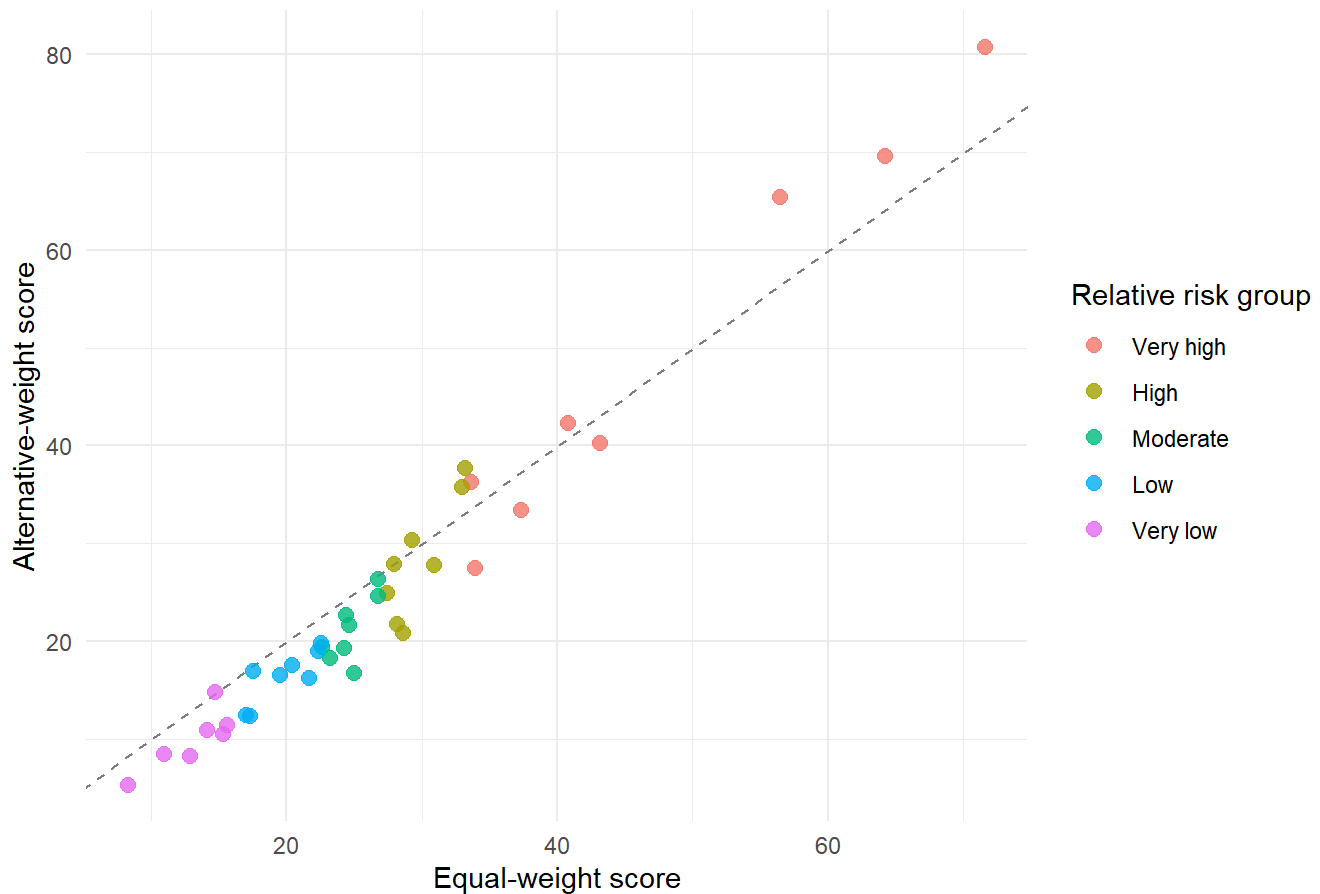
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Sensitivity of the district ranking to alternative weights

measure	value
Spearman rank correlation	0.965
Districts remaining in the top 10	9.000
Largest absolute rank change	9.000

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Sensitivity of Risk Scores to Alternative Weights



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The sensitivity analysis indicates whether the ranking is stable when greater importance is assigned to human and structural consequences. Nevertheless, the score remains a decision-support index based on the available scenario data, not an official earthquake-risk classification. Expert consultation would be required before using the weights for operational resource allocation.

3.3 Urban Transformation Opinions and Risk

The urban transformation variable represents the proportion of respondents in each district who reported that their household needed urban transformation. It is compared below with model-based risk indicators derived from the earthquake scenario, building stock, infrastructure, and population datasets. These indicators are objective in the sense that they are calculated from administrative and scenario data, but they should not be interpreted as observed post-earthquake damage.

3.3.1 Association with Risk Indicators

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Relationship between urban transformation opinions and risk indicators

risk_indicator

spearman_rho

risk_indicator	spearman_rho
Pre-1980 building share	0.554
Structural vulnerability	0.524
Human consequences	0.362
Overall risk score	0.339
Heavy damage share	0.152
Exposure	0.007
Infrastructure and shelter	-0.087

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Spearman test for overall risk and perceived transformation need

measure	value
Spearman correlation	0.339
p-value	0.035
Districts	39.000

Perceived urban transformation need has a positive but modest relationship with the overall risk score ($\rho = 0.339$, $p = 0.035$). The relationship is stronger for the pre-1980 building share ($\rho = 0.554$) and structural vulnerability component ($\rho = 0.524$). In contrast, perceived need is almost unrelated to exposure and infrastructure pressure. Residents' opinions therefore appear to align more closely with visible building age than with the complete multidimensional risk profile.

3.3.2 Risk-Demand Policy Groups

Districts were divided using the median overall risk score and median urban transformation-need proportion. This creates four relative policy groups. The "high risk / low demand" group is particularly important because the modeled risk is above the Istanbul median while perceived transformation need is below the median.

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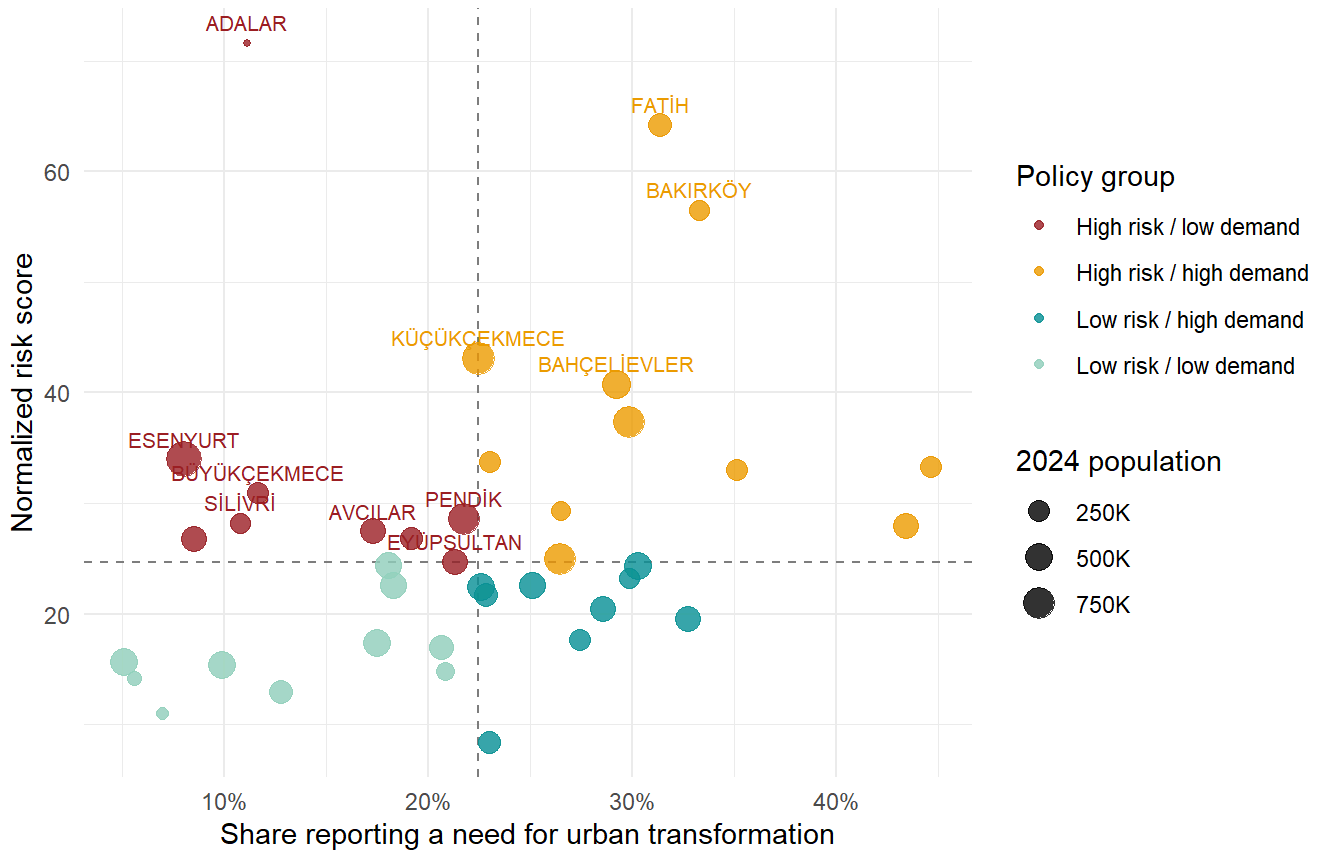
Distribution of districts across risk-demand policy groups

politika_grubu	districts
High risk / low demand	9
High risk / high demand	11
Low risk / high demand	9
Low risk / low demand	10

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Modeled Risk and Perceived Urban Transformation Need

Dashed lines show the median values across 39 districts



Nine districts fall into the high-risk/low-demand group. This does not prove that residents are unaware of risk: the opinion data may reflect financial constraints, tenure status, trust, previous transformation experience, or differences in how the survey question was understood. It does, however, identify districts where technical assessment and public perception are less closely aligned.

3.3.3 Priority Gaps

The risk-demand gap compares each district's percentile position in the risk ranking with its percentile position in transformation demand. Positive values indicate that a district ranks higher in modeled risk than in perceived transformation need. This is a relative prioritization measure, not an estimate of individual awareness.

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Districts with above-median risk and below-median transformation demand

district	risk_rank	risk_score	transformation_need	structural_score	pre_1980_share
ADALAR	1	71.6	11.1%	88.2	56.1%
ESENYURT	7	33.9	8.0%	16.2	1.3%
BÜYÜKÇEKMECE	11	30.9	11.7%	19.8	4.2%
PENDİK	13	28.6	21.8%	14.3	13.0%

district	risk_rank	risk_score	transformation_need	structural_score	pre_1980_share
SİLİVRİ	14	28.2	10.8%	12.2	2.7%
AVCILAR	16	27.4	17.3%	20.7	9.7%
TUZLA	17	26.8	19.1%	23.3	10.2%
BEYLİKDÜZÜ	18	26.8	8.5%	26.8	0.3%
EYÜPSULTAN	20	24.7	21.3%	31.4	34.8%

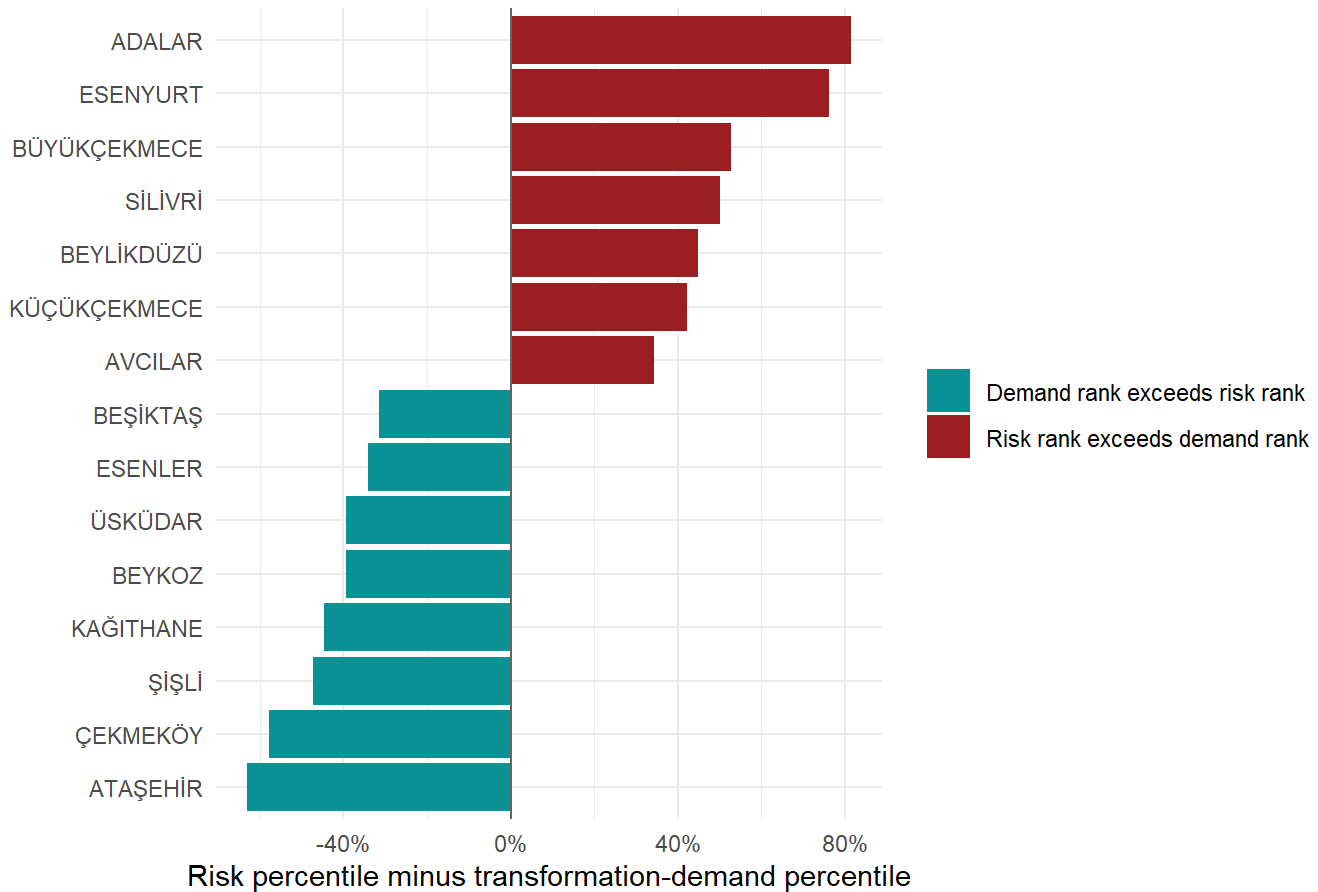
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Largest positive gaps between modeled risk and transformation demand

district	risk_score	transformation_need	percentile_gap
ADALAR	71.6	11.1%	81.6%
ESENYURT	33.9	8.0%	76.3%
BÜYÜKÇEKMECE	30.9	11.7%	52.6%
SİLİVRİ	28.2	10.8%	50.0%
BEYLİKDÜZÜ	26.8	8.5%	44.7%
KÜÇÜKÇEKMECE	43.2	22.4%	42.1%
AVCILAR	27.4	17.3%	34.2%
PENDİK	28.6	21.8%	21.1%
TUZLA	26.8	19.1%	21.1%
BAYRAMPAŞA	33.7	23.1%	21.1%

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Largest Relative Gaps Between Risk and Transformation Demand



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Adalar and Esenyurt have the largest positive gaps. Adalar combines the highest modeled risk score with a comparatively low transformation-demand proportion, while Esenyurt combines substantial exposure with one of the lowest reported demand proportions. Büyükçekmece, Silivri, and Beylikdüzü also show notable positive gaps. These districts are candidates for more detailed local investigation and targeted risk communication rather than automatic conclusions about resident awareness.

3.4 Results Tables, Ranking, and District Map

3.4.1 Priority Summary

The table below combines the principal decision-support indicators. It allows the overall ranking to be interpreted together with projected losses, structural conditions, exposure, and public opinion.

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Decision-support summary for the 15 highest-ranked districts

rank	district	risk_score	risk_group	projected_loss_of_life	loss_per_100000	heavy_damage_share	pre_198
1	ADALAR	71.6	Very high	76	447.6	18.1%	56.1%

rank	district	risk_score	risk_group	projected_loss_of_life	loss_per_100000	heavy_damage_share	pre_198
2	FATİH	64.2	Very high	1484	418.7	12.8%	73.2%
3	BAKIRKÖY	56.5	Very high	1046	475.7	17.5%	40.7%
4	KÜÇÜKÇEKMECE	43.2	Very high	1515	192.0	9.6%	18.8%
5	BAHÇELİEVLER	40.8	Very high	1633	291.6	9.8%	16.7%
6	BAGCILAR	37.4	Very high	1179	165.2	6.2%	14.3%
7	ESENYURT	33.9	Very high	1003	101.5	6.0%	1.3%
8	BAYRAMPAŞA	33.7	Very high	520	193.8	8.6%	39.6%
9	GÜNGÖREN	33.2	High	754	284.7	8.7%	23.0%
10	ZEYTİNBURNU	33.0	High	668	240.0	9.6%	16.2%
11	BÜYÜKÇEKMECE	30.9	High	288	102.7	6.6%	4.2%
12	BEYOĞLU	29.3	High	217	100.1	4.6%	62.4%
13	PENDİK	28.6	High	195	26.0	2.6%	13.0%
14	SİLİVRİ	28.2	High	58	25.0	4.3%	2.7%
15	ESENLER	28.0	High	638	150.6	5.1%	27.7%

The ranking must be interpreted through its components. For example, Adalar ranks first because of very high per-capita consequences despite low total exposure, whereas Küçükçekmece and Bahçelievler combine substantial projected consequences with much larger populations. The table therefore presents both absolute and population-adjusted outcomes.

3.4.2 Complete District Ranking

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Complete normalized risk ranking of Istanbul’s 39 districts

rank	district	risk_score	risk_group	human_consequences	structural_vulnerability	infrastructure_shelter
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rank	district	risk_score	risk_group	human_consequences	structural_vulnerability	infrastructure_shelter
1	ADALAR	71.6	Very high	98.0	88.2	100.0
2	FATİH	64.2	Very high	83.8	84.9	30.6
3	BAKIRKÖY	56.5	Very high	89.9	76.0	43.6
4	KÜÇÜKÇEKMECE	43.2	Very high	37.8	38.4	20.2
5	BAHÇELİEVLER	40.8	Very high	54.8	37.6	24.4
6	BAĞCILAR	37.4	Very high	31.6	25.5	17.6
7	ESENYURT	33.9	Very high	19.6	16.2	15.1
8	BAYRAMPAŞA	33.7	Very high	38.9	49.7	17.4
9	GÜNGÖREN	33.2	High	53.8	38.8	22.6
10	ZEYTİNBURNU	33.0	High	45.8	36.6	25.5
11	BÜYÜKÇEKMECE	30.9	High	20.5	19.8	42.7
12	BEYOĞLU	29.3	High	20.8	53.9	9.7
13	PENDİK	28.6	High	5.5	14.3	8.1
14	SİLİVRİ	28.2	High	5.6	12.2	36.8
15	ESENLER	28.0	High	28.6	31.6	12.0
16	AVCILAR	27.4	High	20.6	20.7	24.6
17	TUZLA	26.8	Moderate	18.4	23.3	27.1
18	BEYLİKDÜZÜ	26.8	Moderate	24.2	26.8	28.7
19	ÜMRANİYE	25.0	Moderate	1.5	8.6	3.2
20	EYÜPSULTAN	24.7	Moderate	8.5	31.4	7.8

rank	district	risk_score	risk_group	human_consequences	structural_vulnerability	infrastructure_shelter
21	KADIKÖY	24.4	Moderate	8.1	41.0	5.4
22	ÜSKÜDAR	24.3	Moderate	3.9	25.2	4.3
23	BEYKOZ	23.3	Moderate	2.5	22.5	7.8
24	KARTAL	22.6	Moderate	7.3	26.7	7.4
25	GAZİOSMANPAŞA	22.6	Low	5.9	33.2	2.7
26	MALTEPE	22.4	Low	8.9	22.9	7.4
27	SARIYER	21.7	Low	2.3	17.2	3.9
28	KAĞITHANE	20.4	Low	3.9	29.5	1.9
29	ATAŞEHİR	19.6	Low	4.6	23.4	6.8
30	ŞİŞLİ	17.6	Low	4.1	37.0	1.1
31	SULTANGAZİ	17.4	Low	2.3	9.7	1.1
32	SULTANBEYLİ	17.0	Low	4.5	5.3	10.4
33	BAŞAKŞEHİR	15.7	Very low	3.2	7.8	4.7
34	SANCAKTEPE	15.4	Very low	2.2	3.8	4.6
35	BEŞİKTAŞ	14.8	Very low	3.4	33.6	4.0
36	ÇATALCA	14.2	Very low	1.4	6.5	22.9
37	ARNAVUTKÖY	12.9	Very low	0.3	0.8	6.0
38	ŞİLE	11.0	Very low	0.0	4.7	22.9
39	ÇEKMEKÖY	8.4	Very low	0.2	1.5	1.3

3.4.3 District Boundary Integration

The district map uses GADM 4.1 second-level administrative boundaries. A fixed lookup based on GADM geographic identifiers is used because several district names in the source file have character-encoding differences. The code stops if the boundary data and risk table do not match exactly.

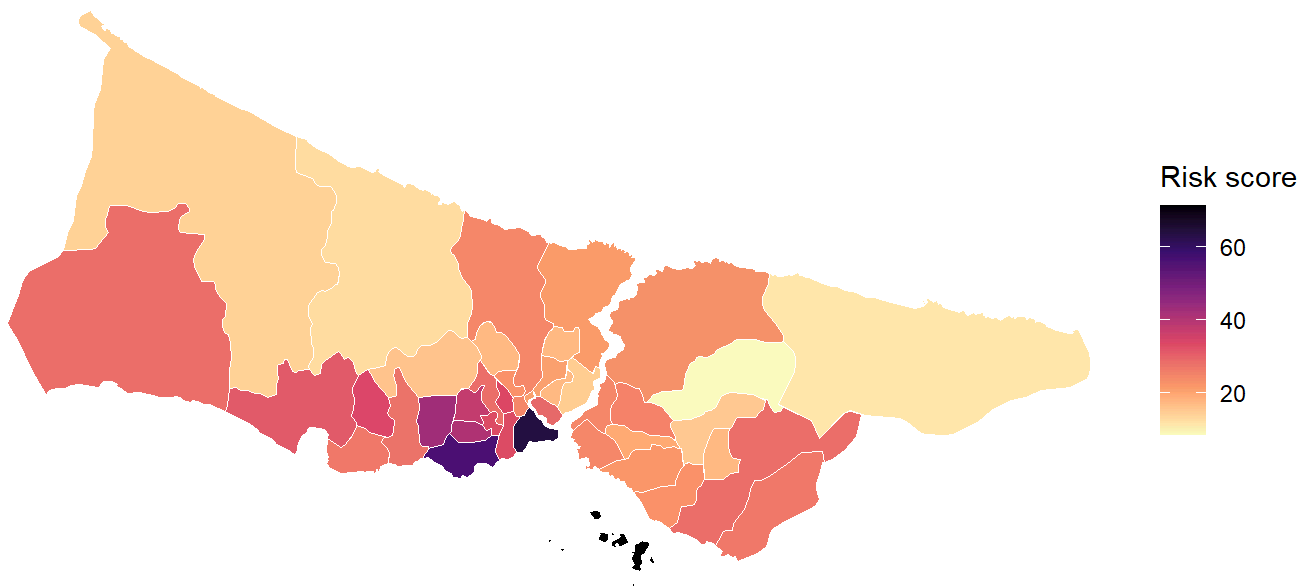
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3.4.4 Static Risk Map

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Normalized Earthquake Risk Score by District

Relative composite score based on four equally weighted dimensions



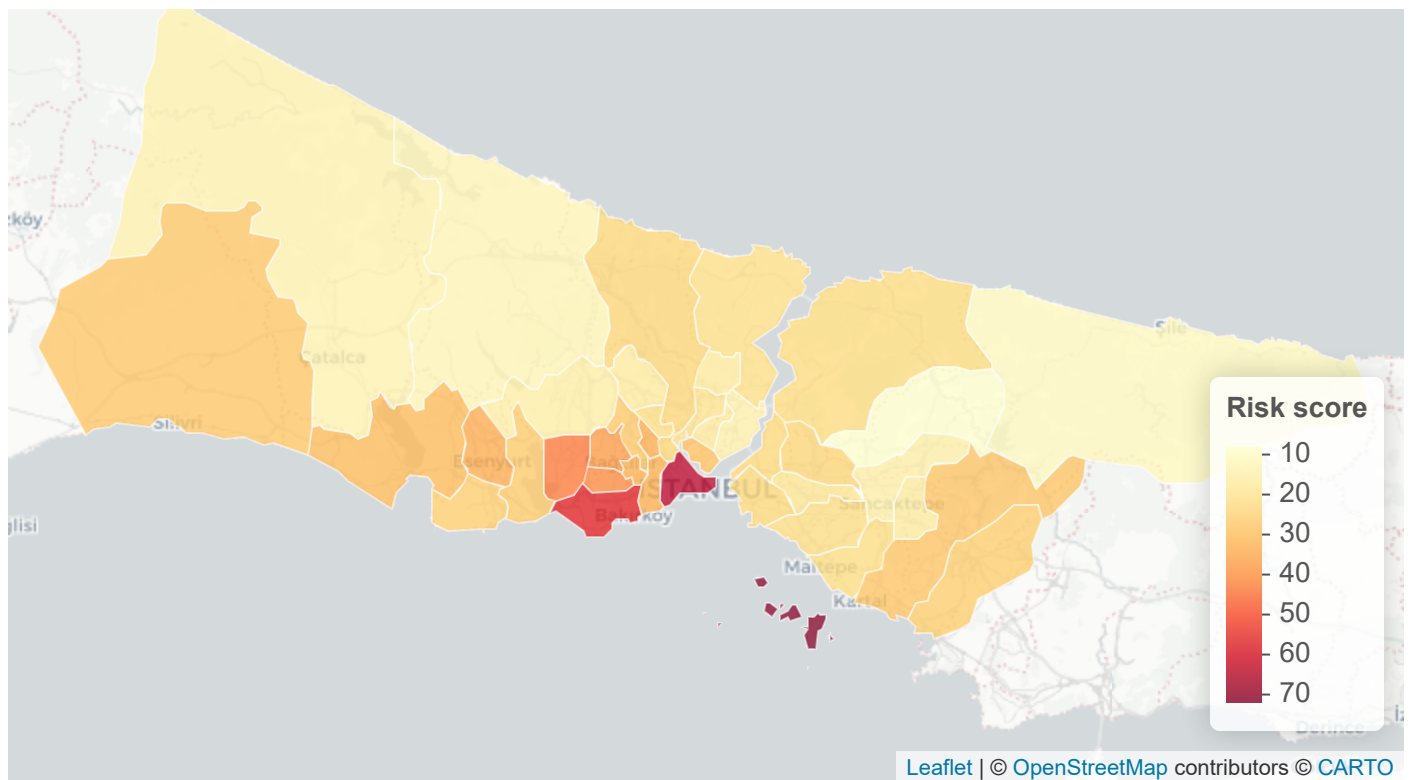
District boundaries: GADM 4.1

The map highlights clusters that are less visible in a simple ranking. High scores appear in both central districts with older building stocks and western districts with high projected consequences or exposure. The map is comparative: lighter districts have lower scores relative to other Istanbul districts, not zero earthquake risk.

3.4.5 Interactive District Map

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3.5 Results

The equal-weight composite score ranks Adalar, Fatih, Bakırköy, Küçükçekmece, and Bahçelievler as the five highest-risk districts. These districts reach high scores for different reasons. Adalar has the highest per-capita human-consequence and infrastructure scores despite having the lowest exposure score, while Fatih combines high human consequences with substantial structural vulnerability. Küçükçekmece and Bahçelievler have lower structural scores than Fatih but much larger exposed populations and building stocks.

Adalar’s first-place ranking requires particular care in interpretation. Its small population increases per-capita rates, so its score reflects high modeled consequences relative to population rather than the largest absolute number of affected people. For this reason, the component scores and raw counts should always be considered alongside the overall ranking.

The alternative weighting scheme produces a Spearman rank correlation of 0.965 with the equal-weight ranking, and nine of the ten highest-ranked districts remain in the top ten. This indicates that the broad ranking is relatively stable, although some individual districts change position when greater weight is assigned to human and structural consequences.

4. Results and Key Takeaways

Istanbul is the most populous city in Europe, which means that in the event of a disaster such as an earthquake, both the scale of destruction and the demand for assistance would be extremely high.

The aim of this project is to examine structural vulnerability, make district-based assessments, and evaluate the views of people living in Istanbul.

Five datasets were used in this study, and a series of plots were produced and interpreted in detail.

This study identified the districts that are most likely to experience building and infrastructure damage in the event of an earthquake. In this way, it provides guidance on where aid should be directed, how it should be delivered, and to what extent support may be required.

Finally, this study serves both as a guide and as a means of encouraging society to be better prepared for such a disaster.

More specifically, the study has:

identified the areas most prone to earthquake risk and the districts most vulnerable, visualized these findings on maps, examined the condition of buildings and infrastructure in Istanbul.